# Masahiro Kanai, Ph.D.

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## Education

**Ph.D.** in Biomedical Informatics, Harvard University
2017–2022 Bioinformatics and Integrative Genomics Program, Harvard Medical School | Boston, MA, USA
Thesis *Fine-mapping complex traits in large-scale biobanks across diverse populations*
Advisors Drs. Mark J. Daly and Hilary K. Finucane

**B.S.** in Bioinformatics, Keio University
2013–2017 Department of Biosciences and Informatics | Yokohama, Japan
Thesis *Integrative multi-omics analysis of renal cell carcinoma*
Advisor Dr. Yasubumi Sakakibara

## Academic Appointments

**Instructor in Medicine** Center for Computational and Integrative Biology
2025–present Massachusetts General Hospital and Broad Institute of MIT and Harvard | Boston, MA, USA

**Research Fellow** Center for Computational and Integrative Biology
2022–2025 Massachusetts General Hospital and Broad Institute of MIT and Harvard | Boston, MA, USA
Advisor Dr. Ramnik J. Xavier

### *Other Appointments*

**Research Consultant** Laboratory for Systems Genetics
2026–present RIKEN Center for Integrative Medical Sciences | Yokohama, Japan

**Visiting Researcher** Department of Genome Informatics, Graduate School of Medicine
2024–present The University of Tokyo | Tokyo, Japan

**Visiting Researcher** Institute for Molecular Medicine Finland (FIMM)
2019–present University of Helsinki | Helsinki, Finland

**Visiting Faculty** Department of Statistical Genetics
2022–2025 Osaka University Graduate School of Medicine | Osaka, Japan

**Research Student** Department of Statistical Genetics
2017–2022 Osaka University Graduate School of Medicine | Osaka, Japan
Advisor Dr. Yukinori Okada

**Research Assistant** Laboratory for Statistical Analysis
2015–2019 RIKEN Center for Integrative Medical Sciences | Yokohama, Japan
Advisors Drs. Yoichiro Kamatani and Yukinori Okada

**Technical Assistant** Department of Human Genetics and Disease Diversity
2014–2016 Tokyo Medical and Dental University | Tokyo, Japan
Advisors Drs. Yukinori Okada and Toshihiro Tanaka

## Leadership

**Project Lead** FinnGen immune multiome project
2022–present Population-scale immune multiome atlas of 10 million cells from 1,108 donors (Kanai, M., et al. *Nature*, 2026)

**Writing Group Lead** COVID-19 Host Genetics Initiative
2021–2023 One of the writing group leads for the Initiative's three *Nature* reports (2021, 2022, 2023)

**Core Analysis Team** Global Biobank Meta-analysis Initiative
2020–2022 Core analysis team (Zhou, W., Kanai, M., et al. *Cell Genomics*, 2022); led meta-analysis fine-mapping and its calibration (SLALOM; Kanai, M., et al. *Cell Genomics*, 2022)

## Honors and Awards

2026 **Semifinalist (Postdoctoral)**, Trainee Research Excellence Award, American Society of Human Genetics
2019–2026 Reviewers' Choice Abstract Award, American Society of Human Genetics (2019, 2021, 2022, 2025, 2026)
2024 **Innovators Under 35 Japan**, *MIT Technology Review*
2020 **Finalist (Predoctoral)**, Charles J. Epstein Trainee Award for Excellence in Human Genetics Research, American Society of Human Genetics
2018–2023 Masason Foundation Fellowship
2017–2022 Nakajima Foundation Fellowship
2011 Third Place, Worldwide Finals, Microsoft Imagine Cup Windows 7 Touch Challenge
2010 First Place, National Finals, and Worldwide Finalist, Microsoft Imagine Cup Software Design Competition

### *Certification*

2015 Certified Bioinformatics Engineer, Japanese Society for Bioinformatics

## Professional Service

**Advisory Board** Hail User Advisory Board
2025–present Broad Institute of MIT and Harvard

**Abstract Reviewer** American Society of Human Genetics Annual Meeting
2021, 2024, 2026

**Task Force Member** Polygenic Risk Score Task Force, International Common Disease Alliance
2020–2022 Contributed to the Task Force's perspective on the responsible clinical use of polygenic risk scores (*Nature Medicine*, 2021)

**Ad Hoc Referee** *Advanced Science*; *American Journal of Human Genetics*; *Bioinformatics*; *BMC Bioinformatics*; *Communications Medicine*; *eLife*; *Genetic Epidemiology*; *Genome Biology*; *Human Genetics and Genomics Advances*; *Journal of Human Genetics*; *Journal of the American Heart Association*; *Nature Communications*; *Nature Medicine*; *Nature Neuroscience*; *PLOS Genetics*

**Memberships** American Society of Human Genetics (2016–present); Japan Society of Human Genetics (2014–present); Japanese Society for Bioinformatics (2015–present)

## Publications

First or co-first author on 12 papers (2 as corresponding author); co-senior author on 1. * equal contribution; † corresponding author

### *Selected Publications and Preprints*

1. **Kanai, M.**, Delorey, T. M., Honkanen, J., ..., Graham, D. B., Daly, M. J. & Xavier, R. J. Population-scale immune multiome atlas reveals regulatory disease mechanisms. ***Nature*** (2026). Code Resource Press
   Featured in a Research Briefing of *Nature*
2. Zhang, J., Huang, Y., Claussnitzer, M., ***Kanai, M.** & *Zhou, W. sc-pcQTL: hurdle-based co-expression modeling for multi-gene QTL mapping in single-cell RNA-seq data. ***bioRxiv*** (2026). Code
3. *Karczewski, K. J., *Gupta, R., ***Kanai, M.,** ..., Neale, B. M., Atkinson, E. G. & Martin, A. R. Pan-UK Biobank genome-wide association analyses enhance discovery and resolution of ancestry-enriched effects. ***Nature Genetics* 57,** 2408–2417 (2025). Code Resource
4. **Kanai, M.** & The COVID-19 Host Genetics Initiative. A second update on mapping the human genetic architecture of COVID-19. ***Nature* 621,** E7–E26 (2023). Resource

Lead author of this report; one of the writing group leads for all three of the Initiative's *Nature* reports (2021, 2022, 2023)

5. †**Kanai, M.**, Elzur, R., Zhou, W., Global Biobank Meta-analysis Initiative, Daly, M. J. & †Finucane, H. K. Meta-analysis fine-mapping is often miscalibrated at single-variant resolution. ***Cell Genomics* 2,** 100210 (2022). Code Resource
6. *Weissbrod, O., ***Kanai, M.,** *Shi, H., ..., Martin, A. R., Finucane, H. K. & Price, A. L. Leveraging fine-mapping and multipopulation training data to improve cross-population polygenic risk scores. ***Nature Genetics* 54,** 450–458 (2022). Code Resource
7. †**Kanai, M.**, Ulirsch, J. C., Karjalainen, J., ..., Okada, Y., †Daly, M. J. & †Finucane, H. K. Insights from complex trait fine-mapping across diverse populations. ***medRxiv*** (2021). Code Resource
8. *Sakaue, S., ***Kanai, M.,** Tanigawa, Y., ..., Daly, M. J., Matsuda, K. & Okada, Y. A cross-population atlas of genetic associations for 220 human phenotypes. ***Nature Genetics* 53,** 1415–1424 (2021). Resource Press
Featured on the cover of *Nature Genetics*
9. *Sakaue, S., ***Kanai, M.,** Karjalainen, J., ..., Daly, M. J., Kamatani, Y. & Okada, Y. Trans-biobank analysis with 676,000 individuals elucidates the association of polygenic risk scores of complex traits with human lifespan. ***Nature Medicine* 26,** 542–548 (2020). Press
Featured in News & Views of *Nature Medicine*
10. **Kanai, M.**, Akiyama, M., Takahashi, A., ..., Kubo, M., Okada, Y. & Kamatani, Y. Genetic analysis of quantitative traits in the Japanese population links cell types to complex human diseases. ***Nature Genetics* 50,** 390–400 (2018). Press

## *Other Preprints*

11. Viippola, E., Hartonen, T., Zhu, M., ..., **Kanai, M.**, ..., Okada, Y., Shen, X. & Ganna, A. Sex-related signatures in the plasma proteome not attributable to XX and XY chromosomes and their impact on health and disease. ***medRxiv*** (2026).
12. Liu, Y. C., Cuomo, A. S. E., Huang, Y., ..., **Kanai, M.**, ..., Claussnitzer, M., Neale, B. M. & Zhou, W. Scalable context-dependent single-cell eQTL mapping reveals disease-relevant regulatory variation beyond static models. ***medRxiv*** (2026). Code
13. The ENCODE Project Consortium. The Encyclopedia of DNA elements. ***bioRxiv*** (2026). Resource
14. Palmer, D. S., Hill, B., Hodgson, S., ..., **Kanai, M.**, ..., Karczewski, K. J., Lindgren, C. M. & Neale, B. M. The Biobank Rare Variant consortium powers the discovery of rare genetic associations through global collaboration. ***medRxiv*** (2026). Code Resource
15. Lu, W., Carroll, R. J., Solomonson, M., ..., **Kanai, M.**, ..., Roden, D. M., Neale, B. M. & Karczewski, K. J. Systematic common and rare variant association testing in 392,030 whole genomes in All of Us. ***medRxiv*** (2026). Code Resource
16. Liu, K., Rasmussen, A., Han, W., ..., **Kanai, M.**, ..., Chen, F., Graham, D. B. & Xavier, R. J. Sensitive Transcriptomics and Genotyping reveals function of genetic variants in immunity. ***bioRxiv*** (2025).
17. Finucane, H. K., Parsa, S., Guez, J., **Kanai, M.**, ..., Daly, M. J., Seed, C. & Karczewski, K. J. Variant scoring performance across selection regimes depends on variant-to-gene and gene-to-disease components. ***bioRxiv*** (2024).
18. Koskela, J. T., Häppölä, P., Liu, A., ..., **Kanai, M.**, ..., Laitinen, T., Myllärniemi, M. & Daly, M. J. Genetic variant in SPDL1 reveals novel mechanism linking pulmonary fibrosis risk and cancer protection. ***medRxiv*** (2021).
19. Turley, P., Martin, A. R., Goldman, G., ..., **Kanai, M.**, ..., Daly, M. J., Benjamin, D. J. & Neale, B. M. Multi-Ancestry Meta-Analysis yields novel genetic discoveries and ancestry-specific associations. ***bioRxiv*** (2021). Code

## *Other Publications*

20. Zhou, W., Cuomo, A. S. E., Hu, L., ..., **Kanai, M.**, ..., Powell, J. E., Daly, M. J. & Neale, B. M. Efficient and accurate mixed model association tool for single-cell eQTL analysis. ***Nature Genetics*** (2026). Code Resource
21. Bujnis, M. N., Sterenborg, R. B. T. M., Li, Y., ..., **Kanai, M.**, ..., Jorde, L. B., Medici, M. & Teumer, A. Multi-ancestry genome-wide association analyses provide insights into the genetic basis of Hashimoto's thyroiditis. ***Nature Genetics* 58,** 1855–1865 (2026). Resource
22. Liu, S., Zheng, H., Gu, Y., ..., **Kanai, M.**, ..., Okada, Y., Zhang, Q. & Wei, F. Genome-wide association analyses of gestational phenotypes identify context-specific genetic effects. ***Nature Genetics* 58,** 1845–1854 (2026).
23. Gupta, R., Durham, T. J., Chau, G., **Kanai, M.**, ..., Zhou, W., Neale, B. M. & Mootha, V. K. Mechanism of age-related accumulation of mtDNA mutations in human blood. ***Nature* 656,** 140–148 (2026).
24. Reeve, M. P., **Kanai, M.**, Graham, D. B., ..., Xavier, R. J., Daly, M. J. & Ripatti, S. Genome-wide association analyses of autoimmune hypothyroidism reveal autoimmune and thyroid-specific contributions and an inverse relationship with cancer risk. ***Nature Genetics* 58,** 550–559 (2026).

25. Siraj, L., Castro, R. I., Dewey, H. B., ..., **Kanai, M.**, ..., Reilly, S. K., Ulirsch, J. C. & Tewhey, R. Functional dissection of complex trait variants at single-nucleotide resolution. ***Nature*** (2026). Code Resource
26. Sato, G., Yamamoto, Y., Sonehara, K., ..., **Kanai, M.**, ..., Yamauchi, T., Kadowaki, T. & Okada, Y. Genetic regulation across germline and somatic variation on the Y chromosome contributes to type 2 diabetes. ***Nature Medicine*** **32,** 894–905 (2026).
27. Rossen, J., Shi, H., Strober, B. J., ..., **Kanai, M.**, ..., Liang, L., Weissbrod, O. & Price, A. L. MultiSuSiE improves multi-ancestry fine-mapping in All of Us whole-genome sequencing data. ***Nature Genetics*** **58,** 67–76 (2026). Code Resource
28. Jacobs, H. N., Gorissen, B. L., Guez, J., **Kanai, M.**, ..., Finucane, H. K., Karczewski, K. J. & Burge, C. B. Widespread naturally variable human exons aid genetic interpretation. ***Nature Communications*** **16,** 11345 (2025).
29. Smit, R. A. J., Wade, K. H., Hui, Q., ..., **Kanai, M.**, ..., Sun, Y. V., Berndt, S. I. & Loos, R. J. F. Polygenic prediction of body mass index and obesity through the life course and across ancestries. ***Nature Medicine*** **31,** 3151–3168 (2025).
30. Lammi, V., Nakanishi, T., Jones, S. E., ..., **Kanai, M.**, ..., Marks-Hultström, M., Zeberg, H. & Ollila, H. M. Genome-wide association study of long COVID. ***Nature Genetics*** **57,** 1402–1417 (2025).
31. Sonehara, K., Uwamino, Y., Saiki, R., ..., **Kanai, M.**, ..., Ogawa, S., Okada, Y. & Namkoong, H. Germline variants and mosaic chromosomal alterations affect COVID-19 vaccine immunogenicity. ***Cell Genomics*** **5,** 100783 (2025).
32. Adams, M. J., Streit, F., Meng, X., ..., **Kanai, M.**, ..., Ripke, S., Lewis, C. M. & McIntosh, A. M. Trans-ancestry genome-wide study of depression identifies 697 associations implicating cell types and pharmacotherapies. ***Cell*** **188,** 640–652.e9 (2025).
33. Pozarickij, A., Gan, W., Lin, K., ..., **Kanai, M.**, ..., Millwood, I. Y., Walters, R. G. & China Kadoorie Biobank Collaborative Group. Causal relevance of different blood pressure traits on risk of cardiovascular diseases: GWAS and Mendelian randomisation in 100,000 Chinese adults. ***Nature Communications*** **15,** 6265 (2024).
34. Carey, C. E., Shafee, R., Wedow, R., ..., **Kanai, M.**, ..., Neale, B. M., Walters, R. K. & Robinson, E. B. Principled distillation of UK Biobank phenotype data reveals underlying structure in human variation. ***Nature Human Behaviour*** **8,** 1599–1615 (2024). Code
35. Trsan, T., Peng, V., Krishna, C., ..., **Kanai, M.**, ..., Daly, M. J., Xavier, R. J. & Colonna, M. The centrosomal protein FGFR1OP controls myosin function in murine intestinal epithelial cells. ***Developmental Cell*** **59,** 2460–2476 (2024).
36. Jermy, B., Läll, K., Wolford, B. N., ..., **Kanai, M.**, ..., Mars, N., Ganna, A. & Ripatti, S. A unified framework for estimating country-specific cumulative incidence for 18 diseases stratified by polygenic risk. ***Nature Communications*** **15,** 5007 (2024). Code Resource
37. Sakaue, S., Weinand, K., Isaac, S., ..., **Kanai, M.**, ..., Wei, K., Price, A. L. & Raychaudhuri, S. Tissue-specific enhancer–gene maps from multimodal single-cell data identify causal disease alleles. ***Nature Genetics*** **56,** 615–626 (2024). Code
38. De Vincentis, A., Tavaglione, F., Namba, S., **Kanai, M.**, ..., Valenti, L., Romeo, S. & Vespasiani-Gentilucci, U. Poor accuracy and sustainability of the first-step FIB4 EASL pathway for stratifying steatotic liver disease risk in the general population. ***Alimentary Pharmacology & Therapeutics*** **59,** 1402–1412 (2024).
39. Lo Faro, V., Bhattacharya, A., Zhou, W., ..., **Kanai, M.**, ..., Joos, K., Cox, N. J. & Hirbo, J. Novel ancestry-specific primary open-angle glaucoma loci and shared biology with vascular mechanisms and cell proliferation. ***Cell Reports Medicine*** **5,** 101430 (2024).
40. Scholz, M., Horn, K., Pott, J., ..., **Kanai, M.**, ..., Köttgen, A., Schlosser, P. & Pattaro, C. X-chromosome and kidney function: evidence from a multi-trait genetic analysis of 908,697 individuals reveals sex-specific and sex-differential findings in genes regulated by androgen response elements. ***Nature Communications*** **15,** 586 (2024).
41. Meng, X., Navoly, G., Giannakopoulou, O., ..., **Kanai, M.**, ..., Gelernter, J., Lewis, C. M. & Kuchenbaecker, K. Multi-ancestry genome-wide association study of major depression aids locus discovery, fine mapping, gene prioritization and causal inference. ***Nature Genetics*** **56,** 222–233 (2024).
42. Chen, S., Francioli, L. C., Goodrich, J. K., ..., **Kanai, M.**, ..., Neale, B. M., MacArthur, D. G. & Karczewski, K. J. A genomic mutational constraint map using variation in 76,156 human genomes. ***Nature*** **625,** 92–100 (2024). Resource
43. Cui, R., Elzur, R. A., **Kanai, M.**, ..., Neale, B. M., Fan, Z. & Finucane, H. K. Improving fine-mapping by modeling infinitesimal effects. ***Nature Genetics*** **56,** 162–169 (2024). Code
44. Kerimov, N., Tambets, R., Hayhurst, J. D., ..., **Kanai, M.**, ..., Mosaku, A., Parkinson, H. & Alasoo, K. eQTL Catalogue 2023: New datasets, X chromosome QTLs, and improved detection and visualisation of transcript-level QTLs. ***PLOS Genetics*** **19,** e1010932 (2023). Resource
45. Wang, Y., **Kanai, M.**, Tan, T., ..., Turley, P., Atkinson, E. G. & Martin, A. R. Polygenic prediction across populations is influenced by ancestry, genetic architecture, and methodology. ***Cell Genomics*** **3,** 100408 (2023).
46. International League Against Epilepsy Consortium on Complex Epilepsies. GWAS meta-analysis of over 29,000 people with epilepsy identifies 26 risk loci and subtype-specific genetic architecture. ***Nature Genetics*** **55,** 1471–1482 (2023).

47. Gupta, R., **Kanai, M.**, Durham, T. J., ..., Calvo, S. E., Neale, B. M. & Mootha, V. K. Nuclear genetic control of mtDNA copy number and heteroplasmy in humans. ***Nature* 620,** 839–848 (2023). Code Press

48. Weeks, E. M., Ulirsch, J. C., Cheng, N. Y., ..., **Kanai, M.**, ..., Lander, E. S., Engreitz, J. M. & Finucane, H. K. Leveraging polygenic enrichments of gene features to predict genes underlying complex traits and diseases. ***Nature Genetics* 55,** 1267–1276 (2023). Code Resource

49. Palmer, D. S., Zhou, W., Abbott, L., ..., **Kanai, M.**, Bloemendal, A. & Neale, B. M. Analysis of genetic dominance in the UK Biobank. ***Science* 379,** 1341–1348 (2023). Code Resource

50. Ogawa, K., Tsoi, L. C., Tanaka, H., **Kanai, M.**, ..., Mochizuki, H., Elder, J. T. & Okada, Y. A cross-trait genetic correlation study identified eight diseases and traits associated with psoriasis. ***Journal of Investigative Dermatology* 143,** 1813–1816.e2 (2023).

51. Rämö, J. T., Kiiskinen, T., Seist, R., ..., **Kanai, M.**, ..., Mäkitie, A., Daly, M. J. & Palotie, A. Genome-wide screen of otosclerosis in population biobanks: 27 loci and shared associations with skeletal structure. ***Nature Communications* 14,** 157 (2023).

52. Kurki, M. I., Karjalainen, J., Palta, P., ..., **Kanai, M.**, ..., Runz, H., Daly, M. J. & Palotie, A. FinnGen provides genetic insights from a well-phenotyped isolated population. ***Nature* 613,** 508–518 (2023). Resource Press

53. Wang, Y., Namba, S., Lopera, E., ..., **Kanai, M.**, ..., Okada, Y., Martin, A. R. & Hirbo, J. Global Biobank analyses provide lessons for developing polygenic risk scores across diverse cohorts. ***Cell Genomics* 3,** 100241 (2023). Resource

54. Kanoni, S., Graham, S. E., Wang, Y., ..., **Kanai, M.**, ..., Willer, C. J., Assimes, T. L. & Peloso, G. M. Implicating genes, pleiotropy, and sexual dimorphism at blood lipid loci through multi-ancestry meta-analysis. ***Genome Biology* 23,** 268 (2022).

55. Tsuo, K., Zhou, W., Wang, Y., **Kanai, M.**, ..., Global Biobank Meta-analysis Initiative, Daly, M. J. & Martin, A. R. Multi-ancestry meta-analysis of asthma identifies novel associations and highlights the value of increased power and diversity. ***Cell Genomics* 2,** 100212 (2022). Resource

56. Zhou, W., **Kanai, M.**, Wu, K.-H. H., ..., Willer, C. J., Daly, M. J. & Neale, B. M. Global Biobank Meta-analysis Initiative: Powering genetic discovery across human disease. ***Cell Genomics* 2,** 100192 (2022). Resource Press

57. Bhattacharya, A., Hirbo, J. B., Zhou, D., ..., **Kanai, M.**, ..., Pasaniuc, B., Gamazon, E. R. & Cox, N. J. Best practices for multi-ancestry, meta-analytic transcriptome-wide association studies: Lessons from the Global Biobank Meta-analysis Initiative. ***Cell Genomics* 2,** 100180 (2022). Resource

58. Yengo, L., Vedantam, S., Marouli, E., ..., **Kanai, M.**, ..., Wood, A. R., Visscher, P. M. & Hirschhorn, J. N. A saturated map of common genetic variants associated with human height. ***Nature* 610,** 704–712 (2022). Press

59. Wang, Q. S., Edahiro, R., Namkoong, H., ..., **Kanai, M.**, ..., Kanai, T., Fukunaga, K. & Okada, Y. The whole blood transcriptional regulation landscape in 465 COVID-19 infected samples from Japan COVID-19 Task Force. ***Nature Communications* 13,** 4830 (2022).

60. Ruotsalainen, S. E., Surakka, I., Mars, N., ..., **Kanai, M.**, ..., Widen, E., Daly, M. J. & Ripatti, S. Inframe insertion and splice site variants in MFGE8 associate with protection against coronary atherosclerosis. ***Communications Biology* 5,** 802 (2022). Press

61. Namkoong, H., Edahiro, R., Takano, T., ..., **Kanai, M.**, ..., Kanai, T., Fukunaga, K. & Okada, Y. DOCK2 is involved in the host genetics and biology of severe COVID-19. ***Nature* 609,** 754–760 (2022).

62. Ramdas, S., Judd, J., Graham, S. E., ..., **Kanai, M.**, ..., Willer, C. J., Zhu, X. & Brown, C. D. A multi-layer functional genomic analysis to understand noncoding genetic variation in lipids. ***The American Journal of Human Genetics* 109,** 1366–1387 (2022).

63. The COVID-19 Host Genetics Initiative. A first update on mapping the human genetic architecture of COVID-19. ***Nature* 608,** E1–E10 (2022). Resource

64. Winkler, T. W., Rasheed, H., Teumer, A., ..., **Kanai, M.**, ..., Köttgen, A., Pattaro, C. & Heid, I. M. Differential and shared genetic effects on kidney function between diabetic and non-diabetic individuals. ***Communications Biology* 5,** 580 (2022).

65. Mars, N., Kerminen, S., Feng, Y.-C. A., **Kanai, M.**, ..., Ganna, A., Martin, A. R. & Ripatti, S. Genome-wide risk prediction of common diseases across ancestries in one million people. ***Cell Genomics* 2,** 100118 (2022). Press

66. Zheng, J., Zhang, Y., Rasheed, H., ..., **Kanai, M.**, ..., Åsvold, B. O., Zhang, H. & Gaunt, T. R. Trans-ethnic Mendelian-randomization study reveals causal relationships between cardiometabolic factors and chronic kidney disease. ***International Journal of Epidemiology* 50,** 1995–2010 (2022).

67. Graham, S. E., Clarke, S. L., Wu, K.-H. H., ..., **Kanai, M.**, ..., Deloukas, P., Sun, Y. V. & Willer, C. J. The power of genetic diversity in genome-wide association studies of lipids. ***Nature* 600,** 675–679 (2021). Resource

68. Polygenic Risk Score Task Force of the International Common Disease Alliance. Responsible use of polygenic risk scores in the clinic: potential benefits, risks and gaps. ***Nature Medicine* 27,** 1876–1884 (2021).

69. Luo, Y., **Kanai, M.**, Choi, W., ..., Han, B., McLaren, P. J. & Raychaudhuri, S. A high-resolution HLA reference panel capturing global population diversity enables multi-ancestry fine-mapping in HIV host response. ***Nature Genetics* 53,** 1504–1516 (2021). Code

70. Griesemer, D., Xue, J. R., Reilly, S. K., ..., **Kanai, M.**, ..., Novina, C. D., Tewhey, R. & Sabeti, P. C. Genome-wide functional screen of 3'UTR variants uncovers causal variants for human disease and evolution. ***Cell* 184,** 5247–5260.e19 (2021).

71. Reilly, S. K., Gosai, S. J., Gutierrez, A., ..., **Kanai, M.**, ..., Finucane, H. K., Sabeti, P. C. & Tewhey, R. Direct characterization of cis-regulatory elements and functional dissection of complex genetic associations using HCR–FlowFISH. ***Nature Genetics* 53,** 1166–1176 (2021). Press

72. The COVID-19 Host Genetics Initiative. Mapping the human genetic architecture of COVID-19. ***Nature* 600,** 472–477 (2021). Resource Press

73. Wang, Q. S., Kelley, D. R., Ulirsch, J., **Kanai, M.**, ..., Ardlie, K. G., MacArthur, D. G. & Finucane, H. K. Leveraging supervised learning for functionally informed fine-mapping of cis-eQTLs identifies an additional 20,913 putative causal eQTLs. ***Nature Communications* 12,** 3394 (2021). Code Resource

74. Nakatochi, M., Toyoda, Y., **Kanai, M.**, ..., Okada, Y., Matsuo, H. & Japan Uric Acid Genomics Consortium (Japan Urate). An X chromosome-wide meta-analysis based on Japanese cohorts revealed that non-autosomal variations are associated with serum urate. ***Rheumatology* 60,** 4430–4432 (2021).

75. Pirastu, N., Cordioli, M., Nandakumar, P., ..., **Kanai, M.**, ..., Nivard, M. G., Perry, J. R. B. & Ganna, A. Genetic analyses identify widespread sex-differential participation bias. ***Nature Genetics* 53,** 663–671 (2021).

76. Shi, H., Gazal, S., **Kanai, M.**, ..., Raychaudhuri, S., Sunyaev, S. R. & Price, A. L. Population-specific causal disease effect sizes in functionally important regions impacted by selection. ***Nature Communications* 12,** 1098 (2021). Code

77. Atkinson, E. G., Maihofer, A. X., **Kanai, M.**, ..., Nievergelt, C. M., Daly, M. J. & Neale, B. M. Tractor uses local ancestry to enable the inclusion of admixed individuals in GWAS and to boost power. ***Nature Genetics* 53,** 195–204 (2021). Code Press

78. Chen, M.-H., Raffield, L. M., Mousas, A., ..., **Kanai, M.**, ..., Reiner, A. P., Auer, P. L. & Lettre, G. Trans-ethnic and Ancestry-Specific Blood-Cell Genetics in 746,667 Individuals from 5 Global Populations. ***Cell* 182,** 1198–1213.e14 (2020). Press

79. Vuckovic, D., Bao, E. L., Akbari, P., ..., **Kanai, M.**, ..., Lettre, G., Sankaran, V. G. & Soranzo, N. The Polygenic and Monogenic Basis of Blood Traits and Diseases. ***Cell* 182,** 1214–1231.e11 (2020). Press

80. Shirai, Y., Honda, S., Ikari, K., **Kanai, M.**, ..., Kumanogoh, A., Harigai, M. & Okada, Y. Association of the RPA3-UMAD1 locus with interstitial lung diseases complicated with rheumatoid arthritis in Japanese. ***Annals of the Rheumatic Diseases* 79,** 1305–1309 (2020). Press

81. Ishigaki, K., Akiyama, M., **Kanai, M.**, ..., Kadowaki, T., Kubo, M. & Kamatani, Y. Large-scale genome-wide association study in a Japanese population identifies novel susceptibility loci across different diseases. ***Nature Genetics* 52,** 669–679 (2020). Press

82. Sakaue, S., Hirata, J., **Kanai, M.**, ..., Murakami, Y., Kamatani, Y. & Okada, Y. Dimensionality reduction reveals fine-scale structure in the Japanese population with consequences for polygenic risk prediction. ***Nature Communications* 11,** 1569 (2020). Code Press

83. Ray, J. P., de Boer, C. G., Fulco, C. P., ..., **Kanai, M.**, ..., Lander, E. S., Regev, A. & Hacohen, N. Prioritizing disease and trait causal variants at the TNFAIP3 locus using functional and genomic features. ***Nature Communications* 11,** 1237 (2020).

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### *Reviews and Commentaries*

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